Hermann Schmalzried

Chemical Kinetics of Solids

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Prof. Dr. Hermann Schmalzried
Institut für Physikalische Chemie
und Elektrochemie
Universität Hannover
Callinstraße 3
D-30167 Hannover

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Preface

Chemistry is concerned with reactions, structures, and properties of matter. The scope of this is immense. Alone the chemistry of the solid state cannot be treated in a single monograph to any depth. The course of processes in space and time, and their rates in terms of state variables is the field of kinetics. The understanding of kinetics in the solid state is the aim of this book.

In contrast to fluids, crystals have a greater number of control parameters: crystal structure, strain and stress, grain boundaries, line defects (dislocations), and the size and shape of crystallites, etc. These are all relevant to kinetics. Treatments that go beyond transport and diffusion in this important field of physical chemistry are scarce.

In the previous monograph *Solid State Reactions* (Verlag Chemie, 1975), I attempted to base the understanding of solid state kinetics on point defect thermodynamics and transport theory. In the meantime, a spectacular progress in experimental (*in-situ*) methodology, the growth of materials science (in which practical needs predominate), and a closer acquaintance of chemists with formal theories of non-equilibrium systems have been observed. The question thus arose: Should there be yet another revision of *Solid State Reactions*, following those of 1978 and 1981, or should a new and more comprehensive monograph be written? The answer is this new book. It stresses a deeper conceptual framework on the one side and the unifying aspects of solid state kinetics, despite their multitude and diversity, on the other side. The growing diversity is reflected in fields such as radiation chemistry and mechanochemistry (tribochemistry), for example.

In order to systematize the multitude of solid state processes and their interactions, it seems more important to shape the physico-chemical concepts for relevant limiting cases than to report on many complex reactions in a qualitative manner. This is also reflected in the preponderance of inorganic systems.

Chemical Kinetics of Solids covers a special part of solid state chemistry and physical chemistry. It has been written for graduate students and researchers who want to understand the physical chemistry of solid state processes in fair depth and to be able to apply the basic ideas to new (practical) situations. *Chemical Kinetics of Solids* requires the standard knowledge of kinetic textbooks and a sufficient chemical thermodynamics background. The fundamental statistical theory underlying the more or less phenomenological approach of this monograph can be found in a recent book by A. R. Allnatt and A. B. Lidiard: *Atomic Transport in Solids*, which complements and deepens the theoretical sections.

A large part of *Chemical Kinetics of Solids* was written while I enjoyed the hospitalities of the Theoretical Chemistry Department at Oxford University,

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the CNRS Bellevue Laboratoire Physique des Materiaux (Meudon, France), and the Department of Physical Chemistry at the Polish Academy of Science (Warsaw). The Volkswagenstiftung made the sabbatical leave possible by a generous stipend. Also, the help of the Fonds der Chemischen Industrie has to be mentioned here with gratitude.

Criticisms, encouragement, and the sharing of ideas and time by many coworkers and friends are gratefully acknowledged. The great influence of the late C. Wagner, and in particular of A. B. Lidiard (Oxford) is profoundly appreciated. B. Baranowski (Warsaw), K. D. Becker (Hannover), P. Haasen (Göttingen), M. Martin (Hannover), and Z. Munir (Davis, Cal.) read parts of the manuscript and gave generous advice and suggestions. The graphic work benefited from the skills of C. Majoni. Last but not least, the book would not have been written without the invaluable help of A. Kühn.

Buntenbock, December 1994 *H. Schmalzried*

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Symbols and Definitions

The following list is meant to 1) compile frequently used symbols and 2) define frequently used quantities. If the same symbol has different meanings, it is stated in the text.

If the vector character of a quantity is stressed, the corresponding symbol is set boldface. If only the absolute value matters, the boldface is omitted. The symbol ˆ over a letter designates a tensor (*e.g.*, $\hat{B}$). A basic concept is the structure element (= SE). A crystal is composed of SE's which are characterized by their chemical identity, their sublattice site, and their electric charge. Regular SE's define the perfect crystal. Irregular SE's are point defects in imperfect crystals and include vacancies or interstitials. The general symbol of a SE is $S_{\varkappa}^{q}$: S denotes the chemical unit (element or molecule), q the electric charge, and $\varkappa$ the sublattice. V denotes a vacant site. Often the electric charge q is referred to the perfect crystal (excess quantity, $\cdot$ = positive, $'$ = negative, $\times$ = neutral). The corresponding SE notation is called the Kroeger-Vink notation. When useful, the IUPAC manual *Quantities, Units and Symbols in Physical Chemistry* was used with respect to the notation. Discrepancies in some instances stem from the fact that different symbols are used in different subject fields.

A	area [cm^2]
a_i	activity of chemical component i ($= e^{(\mu_i-\mu_i^{\circ})/RT}$)
b_i	mobility of particles of sort i ($= \boldsymbol{v}_i/\boldsymbol{K}_i$)
c_i	concentration of component i ($= n_i/V$ [mol/cm^3])
D_i	(self)diffusion coefficient of particles of component i
D_i^*	tracer diffusion coefficient of particles of component i
$\tilde{D}$	(chemical) interdiffusion coefficient
e_o	charge of electron ($= 1.6 \cdot 10^{-19}$ C)
e'	electron in crystal
e_c	electron in a conduction band state
$\bar{E}$	modulus of elasticity, Young's coefficient [Pa]
$\boldsymbol{E}$	electric field vector ($= -\nabla\varphi$)
f_i	activity coefficient of component i ($= a_i/N_i$)

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f	correlation factor, jump efficiency in non-random walk motion
F	Faraday constant (= 96485 Coulombs per equivalent)
F	Helmholtz energy (U–TS)
G	Gibbs energy (H–TS)
G_i	partial molar Gibbs energy of component i ($\equiv \mu_i$)
$\triangle G^0$	standard Gibbs energy change of reaction
G	shear modulus (sometimes $\bar{G}$, to distinguish it from Gibbs energy) [Pa]
g_i	Gibbs energy per particle i ($= G_i / N_0$)
h	Planck constant ($= \hbar \cdot 2\pi = 6.626 \cdot 10^{-34}$ [Js])
$\mathrm{h}^{\bullet}$	electron hole in crystal
$\mathrm{h_v}$	hole in a valence band state
H	enthalpy
I	electric current density ($= j_i \cdot z_i \cdot F$)
j_i	flux density of particles of sort i [mol/cm^2 s]
k	Boltzmann constant ($= 1.38 \cdot 10^{-23}$ [JK^{-1}])
$\vec{k}$	(reaction) rate constant in kinetic rate equations
$\boldsymbol{k}$	wave vector, or $\boldsymbol{k} = \lambda / 2\pi$
$\boldsymbol{K}$	force vector
L_i	transport coefficient, generalized conductance of component i
L_D	Debye-Hückel screening length
m_i	mass of particle i
n	amount of substance (number of moles) [mol]
n_i	number of moles of component i
N_i	fraction (mole, site, number of state) of component i
N_0	Loschmid (Avogadro) number ($= 6.02 \cdot 10^{23}$ mol^{-1})
p_i	partial pressure of component i ($= N_i \cdot P$ in ideal gas)
P	pressure ($= \Sigma\, p_i$)
q	(excess) electric charge, characterizing a SE
R	molar gas constant ($= N_0 \cdot k = 8.314$ [JK^{-1} mol^{-1}])
S	entropy

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SE	structure element
t	time
t_i	transport number of component i ($= \sigma_i / \Sigma\ \sigma_i$)
T	temperature [K]
u_i	electrochemical mobility of i ($= \boldsymbol{v}_i / \boldsymbol{E}$)
U	energy, internal energy
$\triangle U$	voltage (difference, change) ($= \varphi_2 - \varphi_1$)
$\boldsymbol{v}_i$	velocity (average drift) of particles i ($\boldsymbol{v}_i$ = velocity vector)
V	volume (V_m = molar volume)
V_i	partial molar volume of component i
$V_\varkappa^q$	vacancy with charge q on sublattice $\varkappa$ (= SE)
X_i	thermodynamic force on species i
z	number (z_i = number of particles i)
z_i	valence number of particles i
Z	partition function
α, β	phase denotation
γ	surface energy, interface energy
δ	deviation from reference value, normally from stoichiometric composition
ε	relative permittivity (dielectric constant)
ε_o	absolute permittivity of vacuum (dielectric constant) ($= 8.854 \cdot 10^{-14}$ [A s/Vcm])
ξ	space coordinate (ξ_1, ξ_2, ξ_3)
$\triangle \xi$	thickness
η	order parameter
η_i	electrochemical potential ($= \mu_i + z_i \cdot F \cdot \varphi$)
λ	wave length
μ_i	chemical potential of particles i ($\equiv G_i = (\partial G / \partial n_i)_{P,T,n_j \neq n_i} = \mu_i^o + RT \ln a_i$)
μ_i^o	standard value of μ_i, usually referred to pure substance i with $a_i = 1$
ν	frequency, transition rate [s^{-1}], sometimes denoted $\omega / 2\pi$
ν	Poisson ratio in theory of elasticity

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ϱ density of mass, charge, particles, etc., specific quantity

σ electrical conductivity ($= \Sigma\, \sigma_i$)

σ entropy production rate

τ time interval, especially relaxation time

φ electric potential ($\nabla\varphi \equiv -\boldsymbol{E}$)

ω angular frequency

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